Content-based Browsing and Retrieval using Image Emphasis

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Abstract

In this paper, a new approach to image database organization is presented. The proposed method is based on both low-level and semantic features, which are ultimately related to relative image appeal or image emphasis. Given a set of pictures in a group, the emphasis image is the image of most importance that should receive the most attention or special treatment. A system for emphasis image selection has been developed and includes three types of features: objective image quality features, people related, and composition related, which are integrated through a Bayesian network for estimating an emphasis value. The proposed approach for organizing image databases consists of a first stage where images are organized based on low-level features and a second stage where images within each group are ranked in terms of their emphasis values. This approach may be used to organize the database in a hierarchical manner and allows for efficient browsing where the emphasis image selected is used as the group representative when browsing takes place.

Keywords: Database Browsing; Semantic Features; Image Appeal

1. Introduction

Image database organization is useful for efficient browsing and can lead to improved image retrieval. Database indexing may be performed using low-level or semantic features at several levels of a hierarchical browsing system (see [1] for links and references to existing systems). In this paper, we propose a new framework of image database browsing and content description based on image emphasis. Given a set of images in a cluster, the emphasis image is the image of most importance that can be used as the group representative. Emphasis image selection (EIS) refers to the process of automatically selecting an emphasis image relative to the images in its group. The incorporation of image emphasis for database browsing is analogous to selecting the most appealing representative image, as is done by high-level cognitive human processes and subjective judgement. The challenges of such an endeavor include selection of proper features and knowledge integration through a suitable classification engine. A fundamental characteristic of the EIS feature extraction stage is that it includes both low-level and semantic features, as described in Section 2. The classification stage is based on a Bayesian network classifier, which provides a flexible and easily trainable architecture for knowledge integration. The use of image emphasis on image database browsing and retrieval is presented in Section 3. This approach is particularly well-suited for organizing databases of consumer photographic images, that may be used in automatic albuming applications [2].

2. Emphasis Image Selection

The EIS system was motivated by the ground truth collection results presented in [3]. The ground truth results showed that the important EIS features are not strongly related to objective metrics that are traditionally used for image quality evaluation. The majority of strong positive attributes belong to the features that are related to people, image composition and image subject. The following features were selected based on a compromise between feature importance and implementation complexity cost [4]: (a) features based on objective image quality measures: sharpness, colorfulness, format uniqueness; (b) features related to people: skin area, people present, close-up; (c) features related to composition: saliency map size, centrality, compactness and variation. A brief description of the features is given below.

2.1 Objective Features

It is possible to compute a large number of objective features; however, colorfulness was found to be the most important and sharpness was only marginally important.

Colorfulness: A measure of colorfulness was obtained by examining for the presence of high-saturation colors along various hues, using the following:

a) Perform a color transformation from RGB to luminance/chrominance space:

$$L=(R+G+B)/3 \qquad C1=(2G-R-B)/4 \qquad C2=(B-R)/2 \qquad (1)$$

b) Quantize the chrominance (C1, C2) plane to 12 wedges, which are referred to as angular bins.

c) Measure the number of high-saturation pixels that populate each angular bin.

d) Determine the number of active bins, i.e., the ones that have high-saturation pixel population above a threshold.

e) Compute colorfulness as follows:

$$Colorfulness = \min\{1.0, (No.\ active\ bins)/10\} \qquad (2)$$

Note that colorfulness has a maximum value of 1.0 when at least 10 out of the 12 bins are populated.

Sharpness: The sharpness is estimated using the edge profile of the image.

a) Image edges are detected from the smoothed luminance channel using the Sobel operator.

b) The edge histogram is formed and the regions that contain the strongest edges are identified as those that are above the 90th percentile of the edge histogram.
c) Strong-edge regions are refined via median filtering, and the average of the strongest edges provides an estimate of sharpness.

Format Uniqueness: The ground truth experiment indicated that if a picture is the only "Panoramic" picture in a group, it is more likely to be selected as the emphasis image. The relative feature "Format Uniqueness" represents this aspect property.

2.2 People-Related Features

People-related features are extremely important in determining image emphasis, but many of the positive attributes that are related to people are difficult to compute, e.g., people smiling, people facing camera, etc. Skin detection methods [5] allow the computation of some people-related features such as: people present, skin area, and close-up. An example of skin detection is shown in Figure 1, where the skin detection map indicates image regions that have high likelihood of being skin. The skin detection method used in this work will be described in future publications.

Skin Area: The percentage of skin/face area in a picture is computed as a preliminary step for determining the presence of people. Skin area is a continuous variable between 0 and 1 and correlates to a number of features related to people. For example, for pictures taken from the same distance, increasing skin area indicates that there are more people in the picture and correlates with the positive indicator of "whole group in photo." Alternatively, if two pictures contain the same number of people, larger skin area may indicate larger magnification, which correlates with the positive attribute of "close-up." Other explanations for larger skin area are possible due to subject positioning, etc., but they were not addressed at the ground truth study.

People Present: The presence of people is detected when a significant amount of skin area is present in the image. The number of skin pixels in the image is counted and people are assumed present when the skin area is above a threshold. "People present" is a binary feature indicating the presence or absence of people.

Close-up: A simple measure of determining close-up is obtained as follows:

a) Skin detection is performed and the resulting map is examined in the central region (25% from border).
b) Close-up is determined as the percentage of skin area in the central portion of the image.

Face detection [6] is sometimes more appropriate than skin detection for determining close-up, but, computationally expensive face detection methods were not pursued in this work. Face detection will be incorporated in a future version of the EIS system.

2.3 Composition Features

Good composition is the most important positive attribute of picture emphasis and bad composition is the most important negative attribute [2]. Good composition may follow a number of general rules [7], but these rules are often violated in order to express the photographer's perspective. The composition measures considered here are with respect to an image saliency map [8]. The features of "Borderness" and "Centrality" are considered in judging the saliency of image regions to construct a saliency map (see Figure 2). The saliency map is quantized to three levels using k-means clustering of its intensity values. The three regions correspond to most salient regions (white), least salient regions (dark), and intermediate regions (gray). The high-saliency regions in the quantized map are likely to correspond to the main subject in the picture; thus, they provide cues to how well the picture is composed. Using the quantized saliency map, the features of main subject size, centrality, compactness, and variation are computed.

2.4 The Classification Stage

The outputs of the feature extraction stage represent statistical evidences that are integrated by a Bayesian network to compute the belief that the processed image is the emphasis image [4], as shown in Figure 3. After the emphasis values have been computed for all images, the one with the highest emphasis is chosen as the emphasis image. Different evidences may compete or reinforce each other according to knowledge derived from the results of the ground truth study. Competition and reinforcement are resolved by the inference engine of a Bayesian network.

A single-level Bayes net is used in the current system, where the emphasis score is determined at the root node and all the feature detectors are at the leaf nodes. It should be noted that each link is assumed to be conditionally independent of other links at the same level, which results in convenient training of the entire net by training each link separately, i.e., deriving the CPM for a given link independent of others [9]. In practice, this assumption is violated; however, the independence simplification/approximation makes implementation feasible and produces reasonable results. A multi-level Bayesian network for emphasis image selection is currently under development.

3. Hierarchical Database Organization

3.1 Image Organization

Image database organization based on image emphasis is accomplished using a hierarchical tree structure that is constructed as follows.

a. The images in the database are grouped in small clusters (<20 images) using low-level features, such as color and texture. If temporal information is available, it can be used to generate the clusters based on event partitioning [1].
b. Within each image cluster, the images according to emphasis, or relative appeal, using the system described in Section 2.
c. The image with the highest emphasis is used as the representative image in its cluster.
d. Emphasis images, that represent clusters at the same hierarchy level, are grouped to form another level of image clusters.
e. The above steps are repeated until the number of clusters at the top level reaches a desirable limit.

Image browsing is accomplished by previewing the emphasis images representing the image clusters in the hierarchical tree structure. This method of organizing the image database results in efficient browsing and quick extraction of the most interesting images.

3.2 Image Retrieval

The prototype EIS algorithm has demonstrated that it is possible to select an acceptable emphasis image from a group of images [4]. This is feasible by constructing a system that incorporates intermediate-level features, based on the skin detection map and saliency map, that are related to image content, as opposed to low-level features alone [10]. An example of emphasis image selection is shown in Figure 4.

Image retrieval using emphasis is performed as shown in Figure 5. The first stage of retrieval is based on low-level features (color, texture and/or shape). The retrieved images are subsequently ranked in terms of their emphasis value. The retrieved images are presented to the user in order of their emphasis values, which is not necessarily the same as the closest distance in low-level feature space. This new ranking can be used to introduce additional possibilities to the user who may choose to expand the search according to the highest ranked emphasis images.

4. Conclusions

In this paper, a new method for image database organization based on the concept of an emphasis image is presented. This method leads to a hierarchical database structure, which allows efficient browsing. In addition, emphasis processing can be used to rank the group of retrieved images obtained through any image retrieval process for more efficient presentation to the system user.

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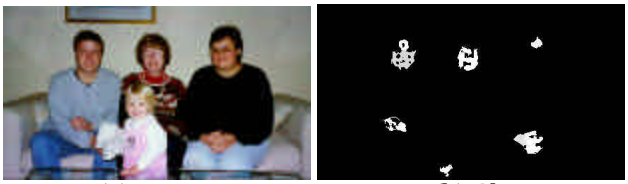
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(a) Image (b) Skin map

Figure 1. Skin detection example.



(a) Image (b) Saliency map (c) Quantized saliency map

Figure 2. Saliency map example.

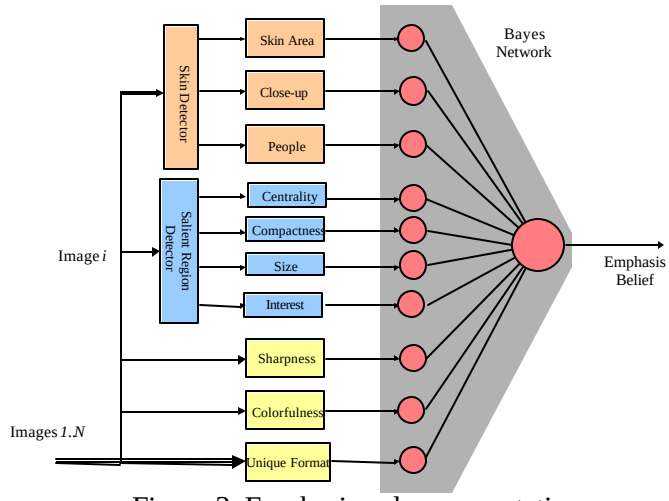


Figure 3. Emphasis value computation.



(a) GT value = 58 (b) GT value = 70 (c) GT value = 93 (d) GT value = 94

Figure 4. Example of EIS results: image (d) was selected; the ground truth results are shown above.

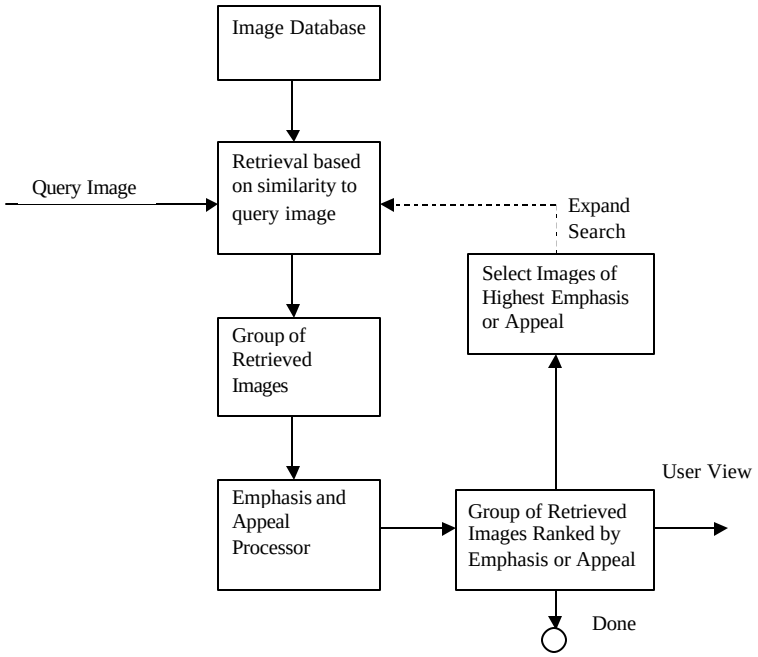


Figure 5. Image retrieval based on emphasis image selection.